

Towards a data-driven circular economy: the potential of predicting material streams in the Finnish construction industry

Joel Himanen

School of Science

Thesis submitted for examination for the degree of Master of Science in Technology.

Espoo 31.12.2022

Supervisor

Prof. Alexander Jung

Advisors

D.Sc. (Tech.) Sami Jokela

Prof. Marko Turpeinen



Aalto University
School of Science

Copyright © 2022 Joel Himanen



Author Joel Himanen

Title Towards a data-driven circular economy: the potential of predicting material streams in the Finnish construction industry

Degree programme Computer, Communication and Information Sciences

Major Machine Learning, Data Science and Artificial Intelligence **Code of major** SCI3044

Supervisor Prof. Alexander Jung

Advisors D.Sc. (Tech.) Sami Jokela, Prof. Marko Turpeinen

Date 31.12.2022

Number of pages 35

Language English

Abstract

Your abstract in English. Keep the abstract short. The abstract explains your research topic, the methods you have used, and the results you obtained.

The abstract text of this thesis is written on the readable abstract page as well as into the pdf file's metadata via the \thesisabstract macro (see above). Write here the text that goes onto the readable abstract page. You can have special characters, linebreaks, and paragraphs here. Otherwise, this abstract text must be identical to the metadata abstract text.

If your abstract does not contain special characters and it does not require paragraphs, you may take advantage of the abstracttext macro (see the comment below).

Keywords Insert, Keywords, Here

Tekijä Joel Himanen

Työn nimi Otsikko suomeksi

Koulutusohjelma Selvitä

Pääaine Selvitä

Pääaineen koodi SCI3044

Työn valvoja Prof. Alexander Jung

Työn ohjaajat TkT Sami Jokela, Prof. Marko Turpeinen

Päivämäärä 31.12.2022

Sivumäärä 35

Kieli Englanti

Tiivistelmä

Tiivistelmässä on lyhyt selvitys kirjoituksen tärkeimmistä sisällöstä: mitä ja miten on tutkittu, sekä mitä tuloksia on saatu.

Avainsanat avainsana1, avainasana2

Preface

Preface.

Otaniemi, 31.12.2022

Joel Himanen

Contents

Abstract	3
Abstract (in Finnish)	4
Preface	5
Contents	6
Abbreviations	7
Acknowledgements	8
1 Introduction	9
2 Background	10
2.1 From linear to circular economies	10
2.2 The role of digitalization, data and machine learning	11
2.3 Circularity in the Finnish construction industry	12
2.4 Motiva's data platform and ecosystem for material circulations	15
2.5 The potential of pre-demolition audits	16
3 The Models	17
3.1 On notation	17
3.2 Random Forest regression	17
3.2.1 Decision trees	17
3.2.2 Forming the ensemble	19
3.3 Bayesian Linear Regression	20
3.3.1 A Bayesian approach to ordinary linear regression	20
3.3.2 Deriving the posterior distribution	21
3.4 Data	22
3.5 Implementation	23
3.5.1 Random Forest	23
3.5.2 Bayesian Linear Regression	23
4 Results	25
4.1 Random forest	25
4.2 Bayesian Linear Regression	25
4.3 Comparison	25
5 Discussion	31
5.1 Interpreting the modeling results	31
5.2 Roadmap for data aggregation and preservation	31
5.3 Future modeling possibilities	32
6 Summary	33

Abbreviations

API	Application programming interface
BLR	Bayesian linear regression
CDW	Construction and demolition waste
CE	Circular economy
GBC	Green Building Council
GHG	Greenhouse gas
HP	Hyperparameter
ML	Machine learning
MVG	Multivariate Gaussian
RF	Random forest

Acknowledgements

1 Introduction

Research questions:

- To what extent is ML-based predictive modeling feasible in the context of CDW streams?
- What does this imply in terms of data aggregation and storage?

2 Background

This chapter establishes the reasoning behind the urgent need for a global transition towards a circular economy (CE) and describes the role digitalization plays in the process. It takes the Finnish construction industry as an example of this transition, and specifies the key challenges and opportunities of driving the change with data and digital solutions.

2.1 From linear to circular economies

It is no news that the ongoing environmental crises, such as global warming and loss of biodiversity, pose a large part of the most significant threats of our lifetime. The challenge of steering humankind towards a sustainable way of life is monumental, and requires extraordinary feats, as well as cross-disciplinary coordination and co-operation from all entities ranging from individual people to businesses, industries, countries and international alliances.

In 2020, the Finnish Innovation Fund Sitra conducted a study on the megatrends for the beginning decade [4], identifying five different, but intertwined global megatrends, three of which are the following:

- *The urgent need for ecological reconstruction*
- *The redefinition of the economy*
- *Technology is embedded in everything*

Getting the interactions between technology and economy to work in favour of fighting environmental crises is an incredibly complex, yet critical challenge. In other words, it is safe to state that the two latter trends play a key role in the way we drive ecological reconstruction in the years to come.

The culprits behind our environmental crises are manifold, but at the end of the day, much of the problem boils down to how we use the natural resources our planet provides us with. Already in 2019, resource extraction and processing made up approximately half of global greenhouse gas (GHG) emissions, in addition to more than 90% of biodiversity loss and water stress. [15] It has become evident, that the sun has set on the "take-make-dispose" linear economy model. We can not afford to extract, use and dispose of resources with the pace we have reached due to assumptions of limitless economic growth with little to no regard to external costs.

The concept of a circular economy presents an alternative model to this, where the use of raw materials and production emissions are decreased by e.g. maximizing product lifetime and reusability, minimizing production waste, and implementing business models based on sharing, reusing and recycling products. [10] Figure 1 summarizes how this kind of circularity is achieved by looping the disposal streams of a linear economy model back to the different stages of the value chain. It is also important to notice the prioritization of those loops; ideally, the inner loops are the most significant ones, since they require less effort and resources to return to the user. For example, it is generally more desirable to return a product to the provider

to be reused as such, than to deconstruct the product, recycle the materials and produce new parts and products out of them.

The potential in this transition is enormous: according to a Material Economics study from 2018 [6], the transition to a more circular economy could reduce GHG emissions from heavy industry in the EU by 56% by 2050. Contrary to a common misconception, this shift does not need to happen at the cost of economic growth as a whole. Aligning technologies and business models to tackle wasteful value creation

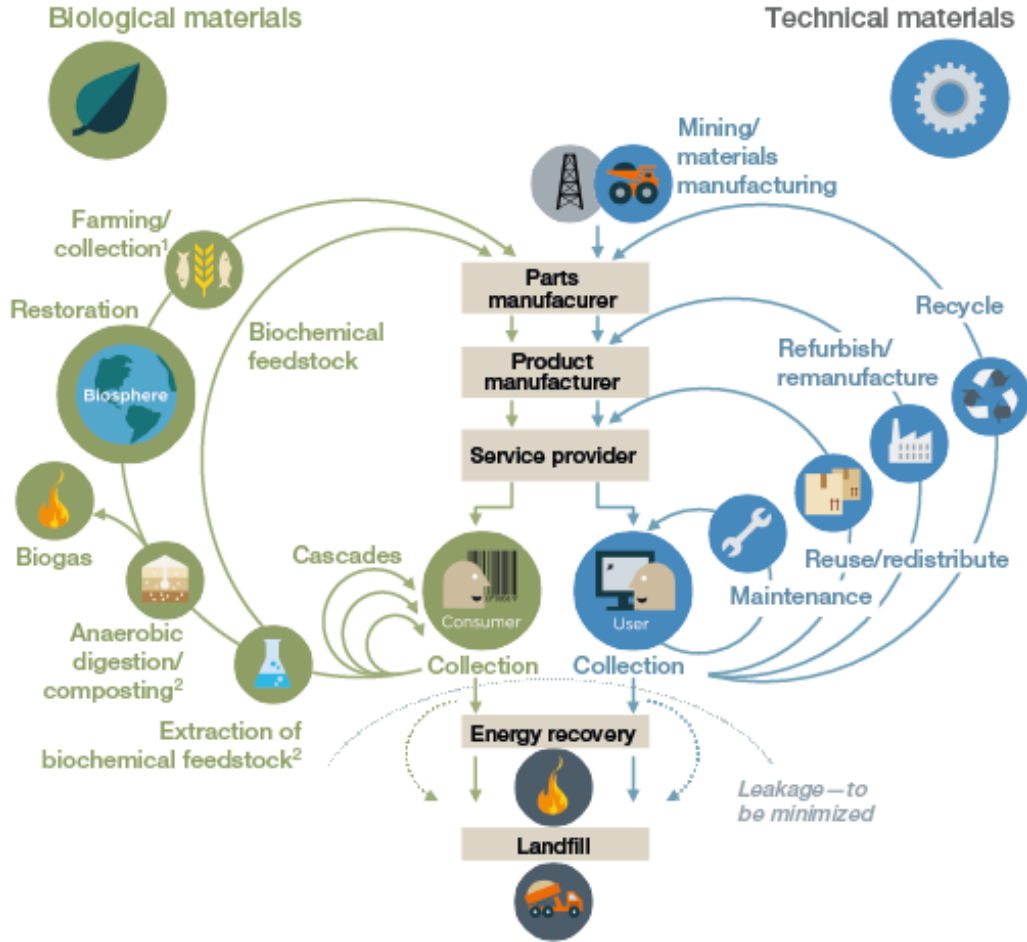


Figure 1: The various loops of a circular economy. *Source: World Economic Forum [23]*

chains and shift towards a more circular model would actually create new markets, and boost employment, lifting EU's GDP by 7% and yielding annual benefits of up to €1,8 trillion by 2030. [7]

2.2 The role of digitalization, data and machine learning

As Sitra states, technology is embedded in everything, and the transition to CE is, of course, no exception. The breathtaking pace digitalization has developed with during the last decades has presented us with countless possibilities and opportunities.

A field that has seen rapid advances in the last decade alone, is Machine learning (ML). Being one of the most widespread modeling paradigms today, ML methods have an impact on all of our lives, whether we want them to or not. Every digital device and transaction generates data, that is utilized in increasing amounts by ML models used to e.g. personalize services and recommendations, and optimize marketing and sales.

In short, the general concept of ML is to take a set of low-level quantities (features) from a data-generating system, and make predictions of quantities of interest (labels) by mapping the features to these labels using a hypothesis map (model). An optimal hypothesis map is obtained from a typically large set of candidate maps by minimizing the discrepancy (loss) between actual label values and those predicted by the hypothesis map candidates. This procedure is referred to as "training" an ML model. [11]

Advances in ML and other areas of data and digitalization, however, come with a significant set of challenges, as well. In addition to a plethora of privacy issues related to emerging data economies, with increasing amounts of digital solutions and services increases also the demand for energy. Hence, it would be naïve to assume digitalization to be a "silver bullet" that directly solves all of our problems. This is also the case in implementing CE models, as is stated in a 2019 study by the European Policy Centre (EPC). [10] That being said, the study focuses on establishing the role of digitalization as a key enabler of CE - as long as it is properly managed. They divide approaches to use digitalization to accelerate the transition to CE into three categories:

1. Improving knowledge, connections and information sharing
2. Making business models, products and processes more circular
3. Strengthening the roles of citizens and consumers

Within the scope of this thesis, we will mainly focus on categories 1 and 2, exploring the ways proper data aggregation and utilization through ML models can support businesses in adapting more circular models. Specifically, we will look at the construction industry, as it is responsible for almost half of global raw material extraction and a quarter of global GHG emissions. [5]

2.3 Circularity in the Finnish construction industry

In order to further clarify the context of this thesis, it is necessary to gain some understanding of the different entities that play a role in the Finnish construction industry, and specifically, what their responsibilities and impacts in terms of construction material circulation is. Figure 2 illustrates how different stakeholders interact in a demolition project. This is, of course, only an example setting and a simplification, but serves to explain the general flow. In some cases, the real estate owner might assume more responsibility and drop the main contractor as a redundant link in the chain. Often it is also the case, that the demolition contractor does the pre-demolition audit as well.

From the point of view of material flow, though, it is imperative to notice that not all materials released during demolition are the responsibility of a single entity. Reusable components and movables, that are possible to extract without heavy machinery, belong to the owner/main contractor, and have to be removed before actual demolition. This process is usually outsourced to recycling operators that

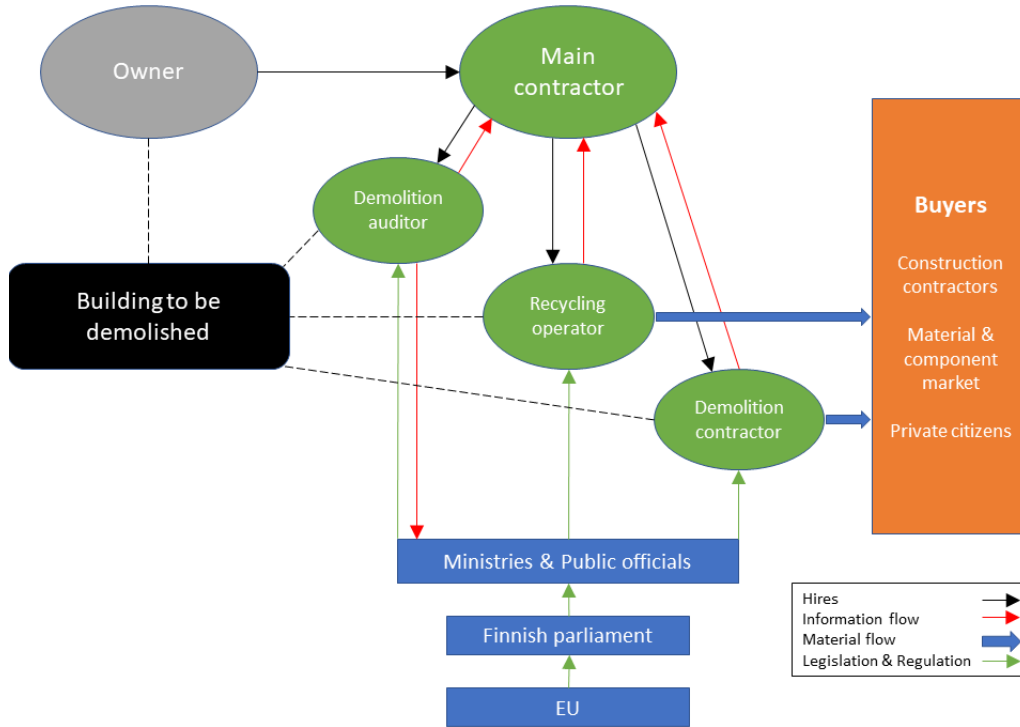


Figure 2: The different roles and their interactions in a generic demolition project.

extract the components and sell them on to new users. Reusable components, that require heavy machinery for extraction, on the other hand, are the responsibility of the demolition contractor, and have to be explicitly specified by the owner/main contractor.

When reviewing CO₂ emissions as part of the environmental impact of the construction industry, it is essential to divide them into three categories according to the phases in the lifecycle of a building:

1. **Upfront carbon** (materials production and construction phase)
2. **Operational carbon** (use phase, energy consumption)
3. **End of life carbon** (demolition phase, demolition waste handling)

These are defined by the World Green Building Council (WorldGBC) in their 2019 report "Bringing embodied carbon upfront" [1]. The same study finds that with the recent advances in the field of renewable energy, operational carbon emissions are reduced significantly. Consequently, upfront and end of life carbon will grow quickly in proportion. In particular, the WorldGBC predicts that upfront carbon will be

responsible for up to half of the whole carbon footprint of new construction between the the present and 2050. A similar trend can be seen in Finland, as well. Up until 2020, approximately 80% of the Finnish built environment's total lifecycle emissions came from operational carbon. [12] With low-carbon energy gaining foothold in the Finnish energy market, focus is shifting increasingly towards reducing upfront and end of life carbon by transitioning to a more circular operating model. The Finnish government has recently taken the lead in driving this transition across industries in Finland. In the spring of 2021, they established a strategic programme [16] to pave the way towards a new Finnish economy based on CE by 2035. The programme dedicates a significant amount of focus on the construction industry, and specifies several action points to promote CE in its context, including financial incentives, digitalization and data management.

From a CE point of view, one of the the main challenges in the Finnish construction industry is construction and demolition waste (CDW) management. Legislation and various costs seem to form a setting, where there is no clear and desirable way for the responsible party to handle CDW. On a demolition site, for example, the prime contractor owns all of the materials and waste (although the responsibility of demolition waste handling lies with the demolition contractor, if the demolition is outsourced by the prime contractor), and therefore is also responsible for processing all of the material streams released during demolition accordingly, including all related costs and taxes. Current legislation makes it difficult, and in many cases expensive, to arbitrarily dispose of CDW. Obviously, this option is also undesirable from an ecological point of view, due to which it is also safe to assume that legislation is going to get even more strict in this matter. Enormous costs and the lack of proper real estate, on the other hand, make it impractical to store CDW even for short spans of time. These matters lead to the conclusion that there is also a significant cost case in addition to the ecological case, that speaks for a circular system where CDW could be salvaged immediately from site for reuse (some components can be extracted pre-demolition and used again as such) and recycling (waste that is not feasible for reuse is processed into new products).

There are, however, significant challenges to overcome. Plenty of costs are embedded in the process of preparing CDW from extraction to reuse and recycling, and said costs often outweigh the costs of disposing of this waste, which makes it less appealing for contractors to salvage it. This problem tends to have the effect of making virgin materials more attractive price-wise, which hinders the demand for reusable and recycled CDW, and makes adapting a more circular model difficult from the point of view of procurement as well. Not only that, but the whole paradigm of construction and most of its processes are built around using virgin materials; a buyer generally risks lower product quality when dealing with reused and recycled materials. This problematic is clearly a key obstacle, as it effects negatively both the supply and demand sides of the non-virgin construction material market. This has also been recognized by the Finnish government, as it has recently set in motion investigation projects to e.g. identify ambiguities in construction product qualification procedures. [24] As an example result, the Finnish Ministry of the Environment released a policy brief [25] on verifying the feasibility of reused materials in June 2022.

It states that it is sufficient to check for feasibility with construction site -specific verification, which means that a CE-marking is not required for e.g. demolition waste that is reused as such.

However, even if there was a strong financial incentive to favor non-virgin materials, a significant challenge remains aligning demand with supply in the reusable and recycled CDW market on a temporal level. Currently, contractors are able to release information of available materials only as soon as they become available, as the demolition project proceeds. Because storage possibilities for these materials are few and costly, the time window to find a buyer is far too narrow. Additionally, this unpredictable batch-like supply leads to discontinuous availability, rather than a continuous supply of non-virgin materials. To counter this challenge, Finnish officials are working on a series of databases and registers that would together provide a centralized data storage system for information on demolition projects and plans, construction product characteristics and material streams, to name a few. These systems are planned to come with diverse APIs to enable private entities to develop their own services and platforms to utilize all of the data stored in these public systems. The following section will introduce one such project, in which the author is also involved through his employment.

2.4 Motiva’s data platform and ecosystem for material circulations

In order to make the raw data from the existing and upcoming registers for construction materials and demolition waste available and usable to companies and consumers, it is vital that different services are developed to refine and process the data. Even though work on the state-wide registers is still in progress, the Finnish state-owned company Motiva Ltd. is already working on a platform for construction material circulation. Motiva is a company of experts specialized in providing organizations and consumers with information, solutions and services on how to use energy and materials more efficiently and sustainably. With this platform, it is Motiva’s aim to build a hub that collects information from various sources under a single platform, acts as a meeting place for supply and demand of non-virgin construction materials and supports all relevant stakeholders in adapting to a more circular operating model in the construction industry.

At the moment, this platform is still in the planning phase. Motiva has developed the concept for it, and it has been validated and further developed using two different projects as case studies. The first one is a demolition project that takes place in Vattuniemi, Helsinki, where 16 buildings are set to be demolished during 2022-23. The other site is located in the Tampere area, where especially the reuse and circulation of ashes as a construction product in infrastructure building was explored. The latter project ended in Spring 2022 for the time being. These projects are vital in gathering field experience on the challenges and requirements of material circulation. Specifically, Motiva seeks to pilot planned features of their platform, and to gain knowledge on how exactly predictive, data-driven demolition planning supports a circular economy.

2.5 The potential of pre-demolition audits

A critical part of shifting towards predictive demolition planning are the procedures that guide the mapping of the demolition site. These include mapping of harmful, but also reusable and recyclable materials that the building to be demolished consists of. Historically, only the mapping of asbestos and other harmful materials has been required by law, but nowadays it is increasingly common to do a broader material mapping. The most extensive mapping process is the pre-demolition audit (in Finnish "purkukartoitus"), which also includes, for example, reusable movables and recommendations on how to reuse, recycle or dispose of the material streams released during demolition. [13] These audits are at the moment voluntary, but officials are constantly planning ways to encourage real estate owners and contractors to use them as a tool to increase material circulation. This is slightly challenging, since a more extensive audit is also more expensive to the client. It is, however, crucial from the point of view of CE, that pre-demolition audits are carried out as carefully as possible, because gaining knowledge of future material streams is necessary in order to align demand with supply in the non-virgin material market.

Carrying out pre-demolition audits, and especially estimating the amounts of different materials released, is hard to do accurately. All experts interviewed for this thesis agreed on this, and stated several possible reasons for the significant errors that occur even in state-of-the-art audits. Some of the material amounts are especially hard to predict, because they might often be hidden from plain sight in the foundations of the building in question. An example of this kind of material is mineral wool, which is often used as an insulator. Especially in the case of old buildings, where the quality of material documentation might be suboptimal, these kind of "hidden" materials can be nigh unto impossible to predict. Extensive, on-site analyses might yield better results, but these are mostly impractical, because the target buildings are often still in use at the time of conducting the audit. It might also happen, that some renovations or other changes during the building's lifetime have not been properly documented, which causes some material streams to come as a surprise during demolition. Finally, it is also natural that clients want to minimize the costs of a demolition project, and the pre-demolition auditors may have to compromise extensiveness and accuracy of the audit in order to stay within the budget.

Given the potential of pre-demolition audits from a CE point of view, and the nature of the challenges described above, it is evident that making the audit process more data-driven holds significant possibilities to develop construction material circulation. The obvious bottleneck at the moment, however, is the lack of data. There are no standardized, mandatory procedures to store pre-demolition audit, material mapping, or realized material stream data. This leads to the relatively small amount of feasible data to be scattered all over different stakeholders, and to have little to no quality guarantees. This challenge is at least partly being addressed by the projects described in sections 2.3 and 2.4, but this thesis will motivate and set a roadmap for proper data aggregation, utilization and sharing in this industry by demonstrating predictive capabilities using actual proof-of-concept models.

3 The Models

This chapter introduces two distinct modeling approaches, the Random Forest regressor and Bayesian Linear Regression (BLR), to predict demolition material streams using initial knowledge of the target building. It is imperative to note, that these models are only a proof of concept - they are meant to demonstrate that given proper data, it is possible to use various machine learning algorithms to make feasible predictions on different material streams, and thus support the pre-demolition audit process. Hence, these models as such should not be relied on in actual cases.

3.1 On notation

Before applying the models on actual data in section 3.4, we will consider a general supervised learning setting with a dataset $\mathcal{D} = \{\mathbf{z}_1, \mathbf{z}_2, \dots, \mathbf{z}_m\}$, where a vector $\mathbf{z}_i$ denotes a data point consisting of features and labels, and m is the size of the dataset. The feature space $\mathcal{X}$ encloses all possible feature values, and similarly, the label space $\mathcal{Y}$ is the space of all possible label values. In the case of $\mathcal{Y}$ being discrete, we talk about classification problems. In the problem setting of this thesis however, we have $\mathcal{Y} = \mathbb{R}$, which implies a regression problem.

Let n be the number of features in the dataset. The feature vector $\mathbf{x} = [x_1, x_2, \dots, x_n]$ contains all feature values for a single data point. It is often useful to represent all feature vectors in a $m \times n$ matrix

$$\mathbf{X} = \begin{bmatrix} \mathbf{x}_1 \\ \mathbf{x}_2 \\ \vdots \\ \mathbf{x}_m \end{bmatrix}.$$

The label of the i th data point is denoted by y_i , and by extension, all labels in the dataset are represented in the label vector $\mathbf{y} = [y_1, y_2, \dots, y_m]^T$. This notation is used for *observed* labels, while quantities *predicted* by a model are referred to using $\hat{y}$. In the scope of this thesis, we will only consider a single label at a time in order to avoid higher dimensional label matrices.

3.2 Random Forest regression

Introduced by Breiman (2001) [3], the Random Forest algorithm is a supervised ensemble learning method, that is made up by of a group of decision trees. Decision trees and random forests have remained a popular modeling approach because they imitate human reasoning and decision making, and are thus conceptually intuitive. They can be used for classification and regression tasks alike, but we will focus on the latter because of the nature of our problem setting.

3.2.1 Decision trees

A Decision Tree regressor is a predictor that maps features to the label space by essentially going through a series of if-else-statements. Each of these decision

statements checks, whether or not a specific subset of features fills the decision criteria. This is done until the process arrives at a prediction. It is conventional to represent decision trees as graphs, that start at the top with a (root) decision node, and continue down to further decision nodes, until arriving at a leaf node. Here the decision nodes represent the if-else-clauses, and leaf nodes the predictions. A simple example of such a decision tree for three-dimensional data $\mathbf{z}_i = [x_1^{(i)}, x_2^{(i)}, y^{(i)}]$ is depicted in Figure 3. This tree shatters a training dataset into four subsets, one for

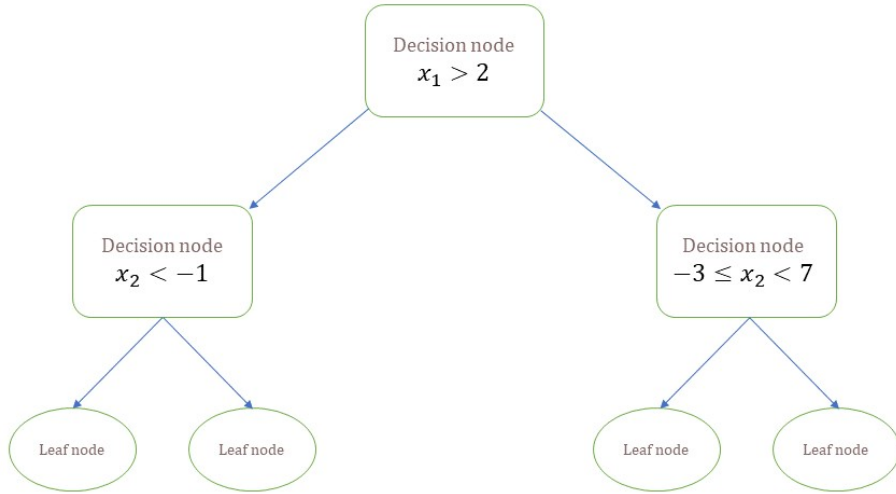


Figure 3: A graph representation of a simple decision tree.

each leaf node. It achieves this by taking one data point at a time, starting at the root node, and choosing the next edge to traverse based on whether the decision statement of the node is true or false in the case of the feature vector in question. If the chosen edge leads to a leaf node, the tree starts to process the next data point. If the next node is a decision node, the decision process is repeated. A new prediction is made similarly by passing the new feature vector through the decision nodes, until it lands on a leaf node. The predicted label is then given a value equal to, for example, the mean of the training labels in the subset associated with that leaf node.

An obvious point of interest remains the question of how to grow an optimal decision tree. There are several possibilities to do this, but we will consider as an example a variant of the ID3 (Iterative Dichotomizer 3) algorithm [18] (Algorithm 1), designed to handle regression problems. The algorithm starts with a training set $\mathcal{D}$ and a root node. It then proceeds to recursively build the two subtrees sprouting from the root node, until maximum tree depth k is reached, or the size of the resulting subset is small enough ($< l$).

The final question to be cleared is the problem of finding the best possible splitting rule, or decision statement, at each decision node. Generally, the metric for the "goodness" of a split is called the *Gain*. Again, there are multiple possibilities when it comes to choosing a Gain function [20], but our choice will be *variance reduction*.

Let $\mathcal{D}_0$ be the dataset at the current node, and $\mathcal{D}_i, \mathcal{D}_j$ be its split, such that $\mathcal{D}_i \cap \mathcal{D}_j = \emptyset$ and $\mathcal{D}_i \cup \mathcal{D}_j = \mathcal{D}_0$. With $\mathcal{Y}$ denoting the set of labels, we can define

the (weighted) variance reduction as

$$\Delta\sigma^2 = \text{Var}(\mathcal{Y}_0) - \left(\frac{|\mathcal{D}_i|}{|\mathcal{D}_0|} \text{Var}(\mathcal{Y}_i) + \frac{|\mathcal{D}_j|}{|\mathcal{D}_0|} \text{Var}(\mathcal{Y}_j) \right). \quad (1)$$

The algorithm goes through every possible split and finally chooses

$$\mathcal{D}_1, \mathcal{D}_2 = \arg \max_{\mathcal{D}_j, \mathcal{D}_j \subseteq \mathcal{D}_0} \Delta\sigma^2$$

to be the respective subsets for the next nodes.

It is worth to mention, that letting the tree grow arbitrarily large results in an infinitely large Vapnik-Chervonenkis dimension for the model, and consequently in unacceptably severe overfitting. [20] Therefore, it is critical to set the bounds k, l

Algorithm 1 ID3 variant for regression

```

1: max_depth  $\leftarrow k$ 
2: min_samples  $\leftarrow l$ 
3: procedure ID3( $\mathcal{D}$ , current_depth = 0)
4:   nr_samples  $\leftarrow |\mathcal{D}|$ 
5:   if current_depth < max_depth & nr_samples > min_samples then
6:      $\mathcal{D}_1, \mathcal{D}_2 \leftarrow \arg \max_{\mathcal{D}_i, \mathcal{D}_j} \text{Gain}(\mathcal{D}, \mathcal{D}_i, \mathcal{D}_j)$ 
7:      $T_L \leftarrow \text{ID3}(\mathcal{D}_1, \text{current\_depth} + 1)$ 
8:      $T_R \leftarrow \text{ID3}(\mathcal{D}_2, \text{current\_depth} + 1)$ 
9:     return Decision node with left and right subtrees  $T_L$  and  $T_R$ 
10:  else
11:    return Leaf node with set  $\mathcal{D}$ 
12:  end if
13: end procedure

```

for tree depth and minimum split size. Optimizing these hyperparameters with more sophisticated methods than simple grid-search is out of the scope of this thesis, though.

3.2.2 Forming the ensemble

Another way to control overfitting and increase robustness of the Decision Tree model is to grow an ensemble of them and choose the average of all decision tree predictions as the final prediction. This method is called the Random Forest. In order to avoid correlated decision trees in the ensemble, the Random Forest algorithm trains each tree using a method called Bagging. [11]

The basic concept is to choose the training set for each tree by uniformly sampling a random subset with replacement from the original dataset. Additionally, the Random forest forces each tree to always consider only a random subset of features when choosing the best split at every decision node (line 6 of Algorithm 1). Introducing these random factors ensures that the generalization error of a random forest converges

as the number of trees in the forest increases [3], which makes it a strong tool to control overfitting.

At this point, the main weakness of the Random Forest becomes evident. Computational complexity quickly becomes a problem, especially as the number of trees increases. Growing a single decision tree, especially with a large training set, is a computationally heavy task due to the need to go through all possible splits at each decision node. This effect is amplified when growing a random forest, resulting in a theoretical computational complexity of $\mathcal{O}(k \cdot t)$ (excluding unification costs) and memory space cost of $\mathcal{O}(2^k)$, where k is the tree depth and t is the number of trees in the forest. [21]

3.3 Bayesian Linear Regression

The second, alternative modeling approach is Bayesian Linear Regression. It is arguably a slightly more advanced method, and offers the means to include domain knowledge in the model in the form of prior distributions, in addition to some metrics for uncertainty in the predictions. Given the scenario this model is applied to in this thesis, one could argue for BLR to be somewhat "overkill" of a choice. However, it goes to demonstrate that also more advanced probabilistic methods are applicable in this context.

3.3.1 A Bayesian approach to ordinary linear regression

We start with the familiar model of ordinary linear regression

$$\mathbf{y} = \mathbf{X}\mathbf{w} + \boldsymbol{\epsilon}, \quad (2)$$

where $\boldsymbol{\epsilon}$ is an error vector. Fitting this model is done by solving the optimization problem of finding the weight vector $\hat{\mathbf{w}}$ that minimizes the sum of squared errors (the so-called Ordinary Least Squares problem)

$$\hat{\mathbf{w}} = \arg \min_{\mathbf{w} \in \mathbb{R}^n} \sum_{i=1}^m |y_i - \mathbf{X}\mathbf{w}|^2 = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y} \quad (3)$$

The Bayesian approach to this model follows the idea that we do not deal with exact values for the vectors $\mathbf{w}$ and $\boldsymbol{\epsilon}$, but rather place some probability distributions on them. Furthermore, we place prior distributions on the parameters of the afore mentioned distributions, which we then update with our data. The updated distribution is called *the posterior*, and it is computed with Bayes' rule

$$p(A|B) = \frac{p(B|A)p(A)}{p(B)}, \quad (4)$$

where $p(A|B)$ is the posterior (the probability of parameter A given the data B), $p(B|A)$ the likelihood (the probability of observing the data B given parameter A), and $p(A)$ is the prior on parameter A . The quantity $p(B)$ is another prior-like

marginal probability, but as we will see in section 3.3.2, it can be dealt with as a normalizing constant when deriving the posterior distribution.

Obtaining the posterior can be cumbersome; at times there might not even be an analytical solution available, and one has to resort to e.g. numerical integration. There is, however, the possibility of using a conjugate prior, which makes computations more straightforward and guarantees the existence of a closed-form posterior. [19] This essentially means, that using a prior that is conjugate to the likelihood results in a posterior of the same family of distributions as the prior.

Going back to the linear model in Equation (2), we now make the assumption that our data is given rise to by a multivariate normal distribution, or Gaussian (MVG). Knowing that the conjugate prior for a MVG likelihood is also a MVG [14], we place on the weights and errors, or noise, the prior distributions

$$\mathbf{w} \sim \mathcal{N}(\boldsymbol{\mu}_0, \alpha^{-1}\mathbf{I}) \quad (5)$$

$$\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \beta^{-1}\mathbf{I}), \quad (6)$$

with $\boldsymbol{\mu}_0, \alpha, \beta$ being hyperparameters interpreted as prior weight means, and weight and noise precision, respectively. We can now write the likelihood as

$$p(\mathbf{y}|\mathbf{w}, \beta) = \mathcal{N}(\mathbf{X}\mathbf{w}, \beta^{-1}\mathbf{I}). \quad (7)$$

3.3.2 Deriving the posterior distribution

Arguably the most challenging part of BLR is coming up with the posterior for $\mathbf{w}$. Even with conjugacy, it still depends on the exact form of the priors and likelihood, and thus has to be derived carefully. Despite the lack of a general formula, we can still usually follow a certain routine to complete the derivation:

1. Start with Bayes' rule (4) and drop all normalizing constant terms independent of $\mathbf{w}$
2. Perform some algebraic manipulations
3. Due to conjugacy, recognize an unnormalized form of a distribution belonging to the same family as the prior

Following these steps, we can now derive the posterior for our case.

Using Bayes' rule (4) and our pre-defined distributions (5)-(7), we can express the (unnormalized) posterior as

$$p(\mathbf{w}|\mathbf{X}, \mathbf{y}, \alpha, \beta) \propto p(\mathbf{y}|\mathbf{w}, \beta)p(\mathbf{w}|\alpha) = \mathcal{N}(\mathbf{y}|\mathbf{X}\mathbf{w}, \beta^{-1}\mathbf{I})\mathcal{N}(\mathbf{w}|\boldsymbol{\mu}_0, \alpha^{-1}\mathbf{I}). \quad (8)$$

Writing out the two Gaussians, dropping constant terms and rearranging, we can

state that the product in (8) is proportional to

$$\begin{aligned} & \exp \left[-\frac{1}{2} \beta (\mathbf{y} - \mathbf{X}\mathbf{w})^T (\mathbf{y} - \mathbf{X}\mathbf{w}) \right] \exp \left[-\frac{1}{2} \alpha (\mathbf{w} - \boldsymbol{\mu}_0)^T (\mathbf{w} - \boldsymbol{\mu}_0) \right] \\ & \propto \exp \left[-\frac{1}{2} \beta \mathbf{w}^T \mathbf{X}^T \mathbf{X} \mathbf{w} + \beta \mathbf{y}^T \mathbf{X} \mathbf{w} - \frac{1}{2} \alpha \mathbf{w}^T \mathbf{w} + \alpha \mathbf{w}^T \boldsymbol{\mu}_0 \right] \\ & = \exp \left[-\frac{1}{2} \mathbf{w}^T \underbrace{(\beta \mathbf{X}^T \mathbf{X} + \alpha \mathbf{I})}_{\boldsymbol{\Sigma}^{-1}} \mathbf{w} + \underbrace{(\beta \mathbf{y}^T \mathbf{X} + \alpha \boldsymbol{\mu}_0^T)}_{\boldsymbol{\mu}^T \boldsymbol{\Sigma}^{-1}} \mathbf{w} \right]. \end{aligned}$$

Completing the square [2], we get that the above expression is proportional to

$$\exp \left[-\frac{1}{2} (\mathbf{w} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1} (\mathbf{w} - \boldsymbol{\mu}) \right],$$

which we recognize as the unnormalized Gaussian $\mathcal{N}(\mathbf{w}|\boldsymbol{\mu}, \boldsymbol{\Sigma})$ with

$$\boldsymbol{\mu} = \boldsymbol{\Sigma} (\beta \mathbf{X}^T \mathbf{y} + \alpha \boldsymbol{\mu}_0) \quad (9)$$

$$\boldsymbol{\Sigma} = (\beta \mathbf{X}^T \mathbf{X} + \alpha \mathbf{I})^{-1}. \quad (10)$$

This concludes the derivation, as we are left with a MVG with mean (9) and covariance (10) as the posterior distribution for the regression weights $\mathbf{w}$.

3.4 Data

The data these models are trained and tested on is based on 16 actual pre-demolition audits carried out on target buildings on the Vattuniemi demo site (2.4). Each building being represented as a data point, the set of features is obtained from initial knowledge of the building. The label set, on the other hand, includes estimates of four different materials released as CDW during demolition, calculated by domain experts in the pre-demolition audit. Some of the feature and label titles, along with an example data point, can be seen below.

Area	Volume	Storeys	Concrete	Mineral wool	Metal	Glass
9590	32400	6	9028.1	20.5	214	25.1

Here, the three chosen features are area, volume, and number of storeys, the units of the former two being m² and m³, respectively. The four labels are measured in metric tons. There are a number of alternative features available, such as categorical variables indicating building type or load bearing material, year of construction, and binary features indicating, whether the target building in question is equipped with an elevator or sewer system. These were, however, left out of this analysis for the sake of computational convenience, as well as the fact that there is little to no variance within the categorical features in this minute dataset.

3.5 Implementation

Both models were implemented in a Python environment, using different libraries for convenience, namely NumPy [9] for computing and pandas [22] for data processing. In order to find optimal hyperparameter (HP) configurations, a grid search (e.g. Algorithm 2) was performed on both models' hyperparameters. In the case of training and test errors, the root-mean-squared error

$$\text{RMSE} = \sqrt{\frac{1}{m} \sum_{i=1}^m (\hat{y}_i - y_i)^2} \quad (11)$$

was used as a metric for the loss between predicted and actual values.

3.5.1 Random Forest

For the Random Forest model, a ready-made implementation in the Scikit-learn library [17] was used, namely `sklearn.ensemble.RandomForestRegressor`. As Algorithm 1 states, the hyperparameters characterizing the forest are maximum tree depth and minimum number of samples to split. Due to the small size of the dataset and number of features, there are not many sensible values for these hyperparameters. Hence, both were given the domain of integers in the interval [2, 5], from which all possible configurations were considered, as per Algorithm 2. To avoid correlation

Algorithm 2 Grid search for hyperparameter optimization

- 1: Define a space of possible values for each HP
 - 2: best error $\leftarrow$ inf
 - 3: **for** every possible HP configuration **do**
 - 4: Fit the model k times with different train-test splits
 - 5: Compute the test error for each fit
 - 6: **if** average test error < best error **then**
 - 7: best error $\leftarrow$ average test error
 - 8: best HP configuration $\leftarrow$ current HP configuration
 - 9: **end if**
 - 10: **end for**
-

between the individual decision trees, only two of the three features were randomly chosen at each split.

3.5.2 Bayesian Linear Regression

In contrast to the Random forest model, no existing implementations were used for BLR, but the model was built "from scratch". Searching for the hyperparameters defined in section 3.3.1 is not as trivial as was the case with the Random forest. It is hard to make any educated guesses on the prior mean vector μ_0 of the regression weights, because the weights are difficult to interpret for the human mind. We can, however, work around this.

Due to the small size and relatively large variance of the dataset, it is reasonable to set weakly to non-informative priors, i.e. not guide the model too much in any direction. This translates into choosing extremely small precisions α, β , or large variances for the weights and errors. This also decreases the effect of $\boldsymbol{\mu}_0$, which we can now set to equal the zero vector $\mathbf{0}$.

For the grid search, the following HP domain was considered:

$$\alpha, \beta \in [10^{-6}, 10^{-4}].$$

The optimization routine also differed from Algorithm 2 in the metric for the "goodness" of a fit. Instead of measuring test error, each fit was judged by how well it explains the training data, i.e. by its log-likelihood. For the exact expression, the equation for log-likelihood with additive Gaussian noise derived by Barber [2]

$$\log(\mathcal{D}|\Gamma) = -\frac{\beta}{2} \sum_{i=1}^m y_i^2 + \mathbf{d}^T \boldsymbol{\Sigma}^{-1} \mathbf{d} - \log \det(\boldsymbol{\Sigma}) + n \log(\alpha) + m \log(\beta) - m \log(2\pi) \quad (12)$$

was used. Here Γ denotes the model parameters, and

$$\mathbf{d} = \beta \sum_{i=1}^m \mathbf{x}_i y_i.$$

4 Results

This chapter summarizes the results of training and testing both the RF and BLR models on the Vattuniemi data described in 3.4. For model performance metrics, training and test errors, as well as the coefficient of determination R^2 were used. Additionally, the difference between training and test error was observed as an indicator of overfitting.

Concerning R^2 , an exact definition has to be supplied in order to avoid ambiguity. Describing how well the model in question is able to explain the data at hand, $R^2 \in [0, 1]$ is usually defined such that $R^2 = 0$ indicates a baseline equivalent to predicting the mean of the data in all cases, and $R^2 = 1$ implies a 100% accuracy, i.e. that the model is able to fully explain the data. To achieve comparability, a definition compatible with Bayesian models was used here. It was suggested by Gelman et. al. [8], and is based on prediction and residual variances:

$$R^2 = \frac{\text{Var}(\hat{\mathbf{y}})}{\text{Var}(\hat{\mathbf{y}}) + \text{Var}(\mathbf{y} - \hat{\mathbf{y}})} \quad (13)$$

4.1 Random forest

For the Random Forest model, grid search yielded a hyperparameter configuration of (2, 2), denoting maximum tree depth and minimum split size, respectively. Due to high result variance with respect to the train-test split, each configuration was tested by averaging over 10 different splits, i.e. $k = 10$ in terms of Algorithm 2, with a 70% to 30% train-test ratio. Using the optimal HP configuration, 100 RF's were trained and tested, using a different split every time. Figure 5 shows the result of six randomly selected runs, with the estimated amount of concrete plotted as a function of target building area (the other two features were ignored here due to visual convenience). Model diagnostics are depicted in Figure 6.

4.2 Bayesian Linear Regression

In the case of BLR, the HP optimization routine resulted in $\alpha = 3.4 \cdot 10^{-5}$ and $\beta = 10^{-4}$. Similarly to RF, each configuration was averaged over $k = 10$ 70% to 30% train-test splits. Results and diagnostics of analogous test runs can be seen in Figures 7 and 8.

4.3 Comparison

A thorough discussion of the above results will follow in 5.1, but let us make a few first remarks here. It is fairly obvious that both models show strong variation in the results across train-test splits. In addition to that, we notice that the test error tends to be greater than the training error for both models (6c and 8c), which indicates overfitting. [11] However, the quantity of most interest is the coefficient of determination R^2 , as it can be viewed as a metric for the "goodness" of a fit.

In order to be able to compare the two models on a single glance, it is useful to summarize the resulting R^2 scores of fitting both models with 100 different splits (6d and 8d) in two separate histograms. These can be observed in Figure 4. Considering that the R^2 values of the Random forests fall between 0.4 and 0.8, while the scores of the BLR fits remain between 0.2 and 0.45, it is safe to conclude, that on this dataset, the Random forest is able to explain the data better than Bayesian linear regression.

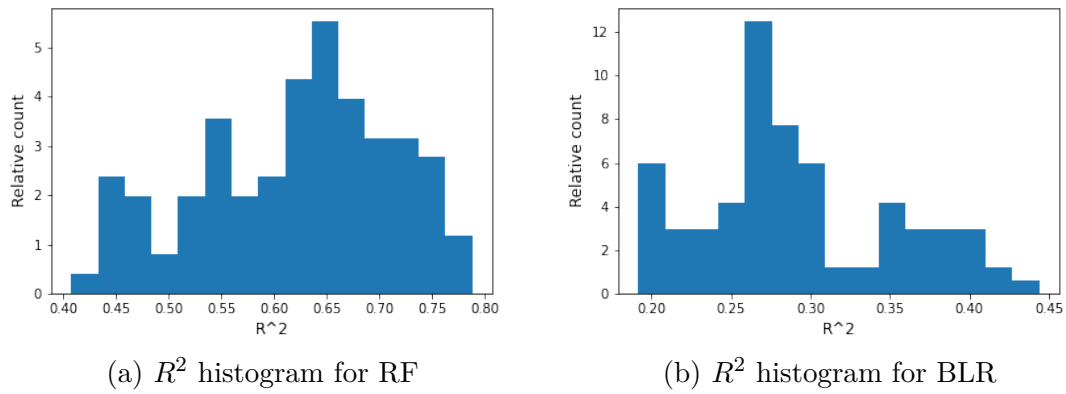
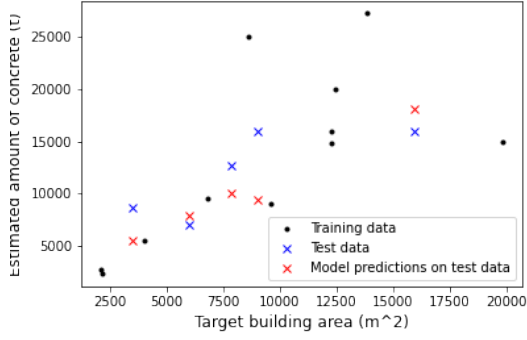
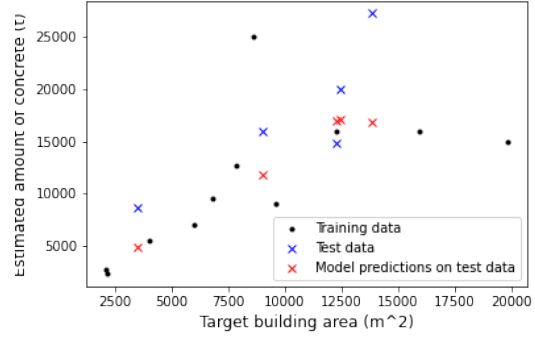


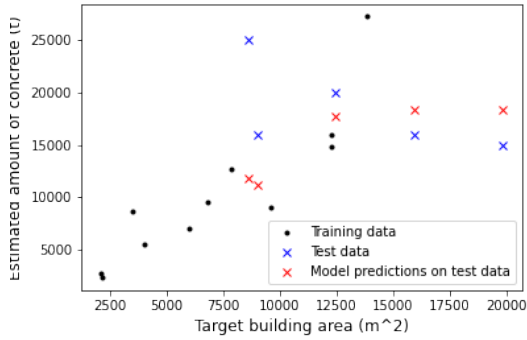
Figure 4: R^2 comparison between RF and BLR



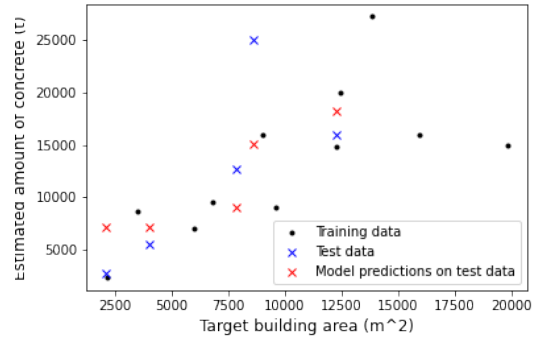
(a) RF test run 1



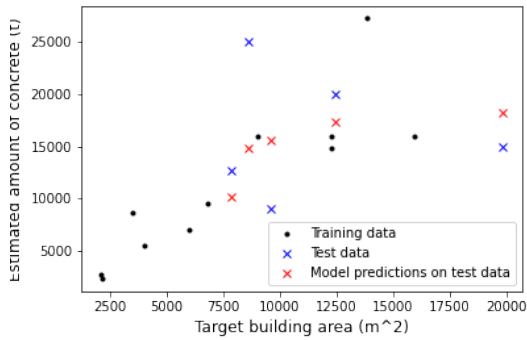
(b) RF test run 2



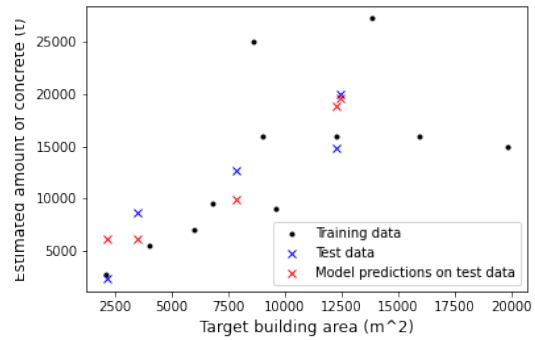
(c) RF test run 3



(d) RF test run 4

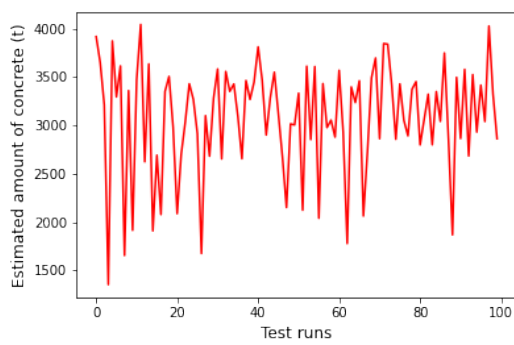


(e) RF test run 5

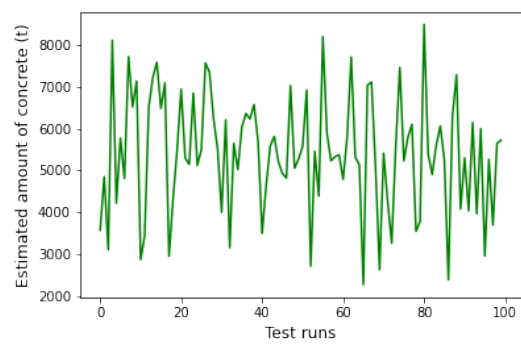


(f) RF test run 6

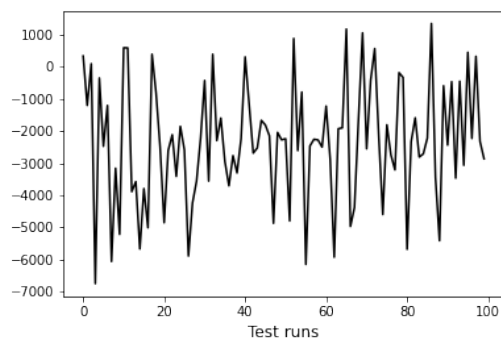
Figure 5: Six randomly selected test runs of the 100 RF models fitted on the Vattuniemi data.



(a) Training errors



(b) Test errors



(c) Difference between training and test errors

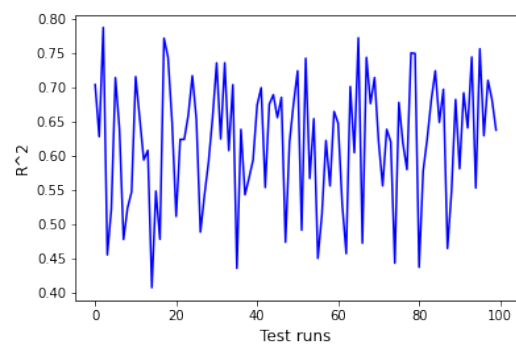
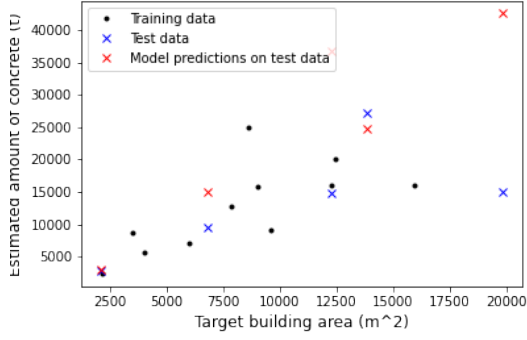
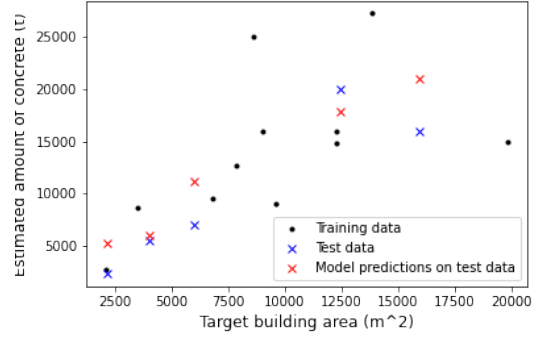
(d) Coefficient of determination R^2

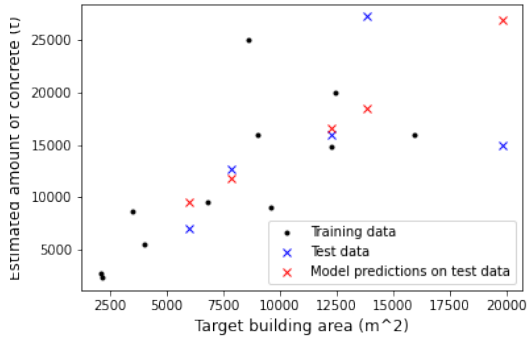
Figure 6: Model diagnostics on the 100 RF test runs



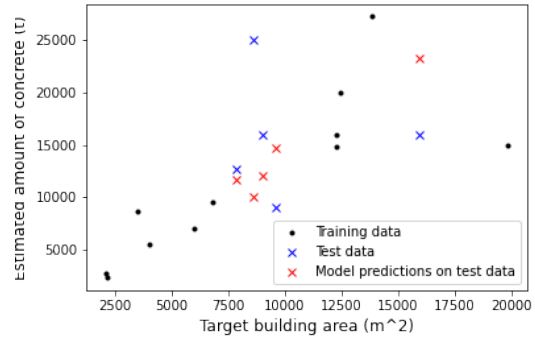
(a) BLR test run 1



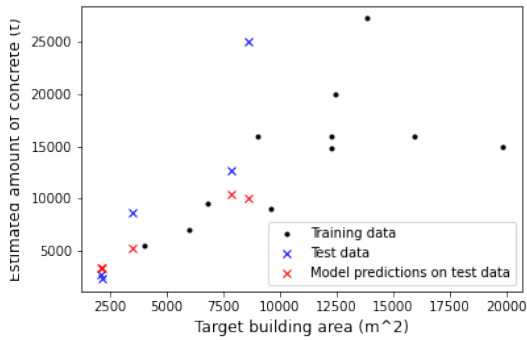
(b) BLR test run 2



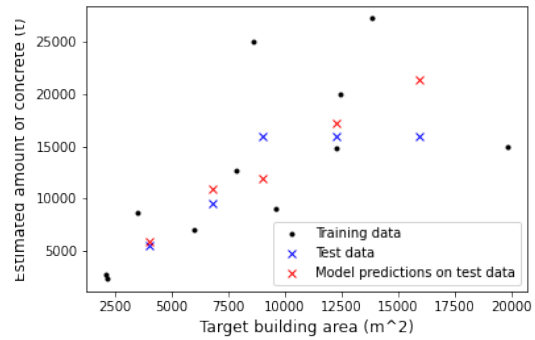
(c) BLR test run 3



(d) BLR test run 4

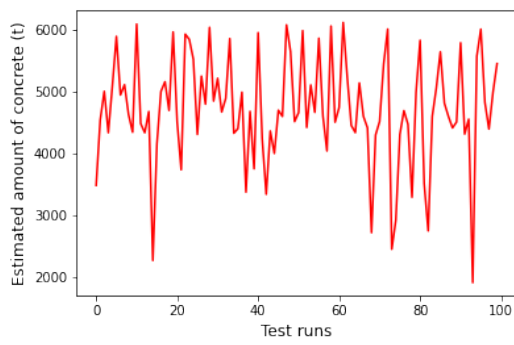


(e) BLR test run 5

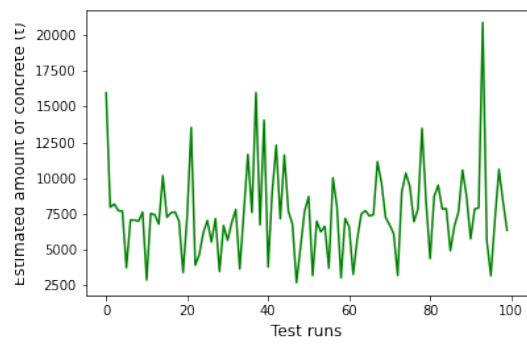


(f) BLR test run 6

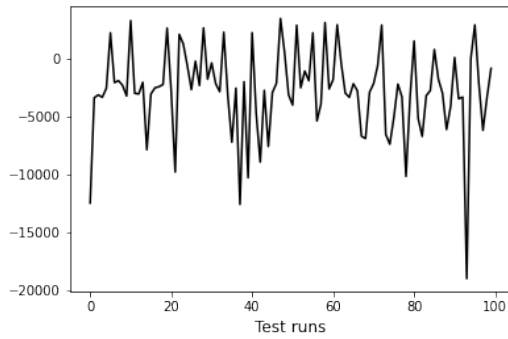
Figure 7: Six randomly selected test runs of the 100 BLR models fitted on the Vattuniemi data.



(a) Training errors



(b) Test errors



(c) Difference between training and test errors

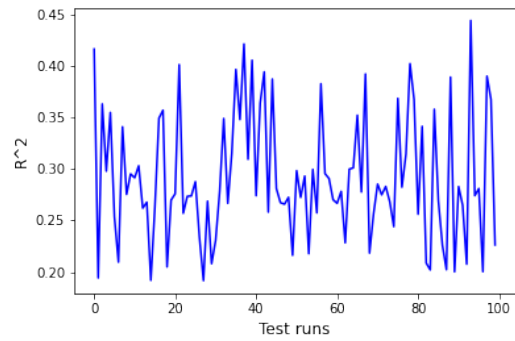
(d) Coefficient of determination R^2

Figure 8: Model diagnostics on the 100 BLR test runs

5 Discussion

5.1 Interpreting the modeling results

5.2 Roadmap for data aggregation and preservation

As was established in section 2.5, there is not much to build on, when it comes to data utilization in the Finnish construction industry. Hence, we have to begin building from the ground up, starting from the question of how to aggregate and store data; predictive modeling only becomes a viable tool when there is enough data to properly train and test the models.

With public entities already working on ways to promote extensive pre-demolition audits, and this study showing that predictive modeling is possible even with minimal demolition material mappings, it is critical to also think about how these documents should be stored in order for them to be easily accessible and usable. The Finnish Ministry of the Environment has already set in motion a project to design and build a public register for CDW data. [LÄHDE?] Preliminary investigations for this register were completed in February 2022, and implementation is scheduled for 2023-24. This register is to serve as a general repository, where contractors and real estate owners submit material mappings and pre-demolition audits.

Up to this point, we have knowingly disregarded a crucial aspect of our data. The proof-of-concept models in Chapter 3 were trained and tested on data based on pre-demolition estimates only. This is, of course, suboptimal. After all, the aim is to predict actual, realized material streams instead of estimates. Data on realized CDW amounts in demolition projects is currently even less available than pre-demolition audits. Therefore, it is crucial that the entities carrying out demolition projects submit detailed reports on actuals, as well as pre-demolition estimates. The previously mentioned public CDW data register would serve as a logical place to store those reports as well.

Considering the data even more closely, it is worth to also question what goes into the pre-demolition audits and actuals. Recalling that one of the most significant obstacles hindering CE in the construction industry is the lack of temporal alignment of demand and supply in the non-virgin material market, it is noteworthy that a temporal aspect in CDW data is currently absent in both pre- and post-demolition reporting. Hence, a key development point in CDW data aggregation is to start including estimates of temporal distributions for the released material streams in pre-demolition audits. Practically, this means stating for each material stream, at which time points during the demolition project a batch of that material would be available, and how large those batches would be. Demolition projects can span several months, so it would be an oversimplification to assume that each material batch would be released as a single batch at the end of the project. Improving these estimates naturally requires that the realized temporal distributions are also properly reported in order to grow a base of quality data for training more sophisticated predictive models.

Another challenge lies with obtaining the features, i.e. basic knowledge of the

target buildings themselves. As stated before, the older end of the Finnish building base is not very well documented, and even in the case of newer buildings, the data is quite scattered and not easily accessible. Streamlining and gathering the data on existing real estate under one roof, and even more importantly, developing the standards for documenting and storing data on new buildings are important factors in making the operating model more circular. It is also crucial not to forget that the structures of buildings change during their lifetime due to e.g. renovations. These changes have to be documented the same as during construction in order to preserve actual information on the building. All in all, it is necessary to aggregate accurate data on buildings during their whole life cycle, and store it in a way that it is accessible through APIs, for example.

5.3 Future modeling possibilities

6 Summary

References

- [1] Matthew Adams, Victoria Burrows, and Stephen Richardson. *Bringing embodied carbon upfront - Coordinated action for the building and construction sector to tackle embodied carbon*. World Green Building Council, 2019.
- [2] D. Barber. *Bayesian Reasoning and Machine Learning*. Cambridge University Press, 2012.
- [3] Leo Breiman. Random forests. *Machine Learning*, 45:5–32, October 2001.
- [4] Mikko Dufva. *Megatrendit 2020*. Sitra, 2020.
- [5] Circle Economy and ABN AMRO. *A futureproof built environment: Putting circular business models into practice*. ABN AMRO, 2017.
- [6] Per-Anders Enkvist and Per Klevnäs. *The Circular Economy - a Powerful Force for Climate Mitigation*. Material Economics, 2018.
- [7] Ellen MacArthur Foundation, McKinsey Center for Business, and Environment. *Growth within: a circular economy vision for a competitive Europe*. Ellen MacArthur Foundation, 2015.
- [8] Andrew Gelman, Ben Goodrich, Jonah Gabry, and Aki Vehtari. R-squared for bayesian regression models. *The American Statistician*, 73(3):307–309, 2019.
- [9] Charles R. Harris, K. Jarrod Millman, Stéfan J. van der Walt, Ralf Gommers, Pauli Virtanen, David Cournapeau, Eric Wieser, Julian Taylor, Sebastian Berg, Nathaniel J. Smith, Robert Kern, Matti Picus, Stephan Hoyer, Marten H. van Kerkwijk, Matthew Brett, Allan Haldane, Jaime Fernández del Río, Mark Wiebe, Pearu Peterson, Pierre Gérard-Marchant, Kevin Sheppard, Tyler Reddy, Warren Weckesser, Hameer Abbasi, Christoph Gohlke, and Travis E. Oliphant. Array programming with NumPy. *Nature*, 585(7825):357–362, September 2020.
- [10] Annika Hedberg and Stefan Šipka. *The circular economy: Going digital*. European Policy Centre, 2020.
- [11] Alexander Jung. *Machine Learning: The Basics*. Springer, Singapore, 2022. <https://alexjungaalto.github.io/MLBasicsBook.pdf>.
- [12] Anna Laine, Tuomas Raivio, Håkan Jonsson, Anna Heino, Markus Klimscheffskij, and Jenny Lehtomäki. *Vähähiilinen rakennusteollisuus 2035 - Osa 1. Rakennetun ympäristön hiilielinkaaren nykytila*. Rakennusteollisuus RT and Gaia Consulting, 2020.
- [13] Katja Lehtonen. *Purkutyöt – opas tekijöille ja teettäjille*. Finnish Ministry of the Environment, 2019.
- [14] Kevin Murphy. Conjugate bayesian analysis of the gaussian distribution. November 2007.

- [15] Bruno Oberle, Stefan Bringezu, Steve Hatfield-Dodds, Stefanie Hellweg, Heinz Schandl, and Jessica Clement. *Global Resources Outlook 2019: Natural Resources for the Future We Want*. United Nations Environment Programme, 2019.
- [16] Ministry of the Environment, Ministry of Economic Affairs, and Employment in Finland. *New directions: The strategic programme to promote a circular economy*. Finnish Government, 2021.
- [17] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, J. Vanderplas, A. Passos, D. Cournapeau, M. Brucher, M. Perrot, and E. Duchesnay. Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12:2825–2830, 2011.
- [18] J.R. Quinlan. Induction of decision trees. *Machine Learning*, 1:81–106, March 1986.
- [19] Howard Raiffa and Robert Schlaifer. *Applied Statistical Decision Theory*. Harvard Business School Publications. Division of Research, Graduate School of Business Administration, Harvard University, 1961.
- [20] Shai Shalev-Shwartz and Shai Ben-David. *Understanding Machine Learning: From Theory to Algorithms*. Cambridge University Press, 2014.
- [21] X. Solé, A. Ramisa, and Carme Torras. Evaluation of random forests on large-scale classification problems using a bag-of-visual-words representation. *Frontiers in Artificial Intelligence and Applications*, 269:273–276, January 2014.
- [22] Wes McKinney. Data Structures for Statistical Computing in Python. In Stéfán van der Walt and Jarrod Millman, editors, *Proceedings of the 9th Python in Science Conference*, pages 56 – 61, 2010.
- [23] McKinsey & Company World Economic Forum, Ellen MacArthur Foundation. *Towards the Circular Economy: Accelerating the scale-up across global supply chains*. World Economic Forum, 2014.
- [24] Ying Zhu, Harriet Lonka, Katja Tähtinen, Markku Anttonen, Päivi Isokääntä, Anssi Knuutila, Jukka Lahdensivu, Selma Mahiout, Anne-Marie Mäntylä, Markku Raimovaara, Tiina Rantio, Tiina Santonen, and Tuuli Teittinen. *Suitability of demolition materials for different applications from the safety and health point of view*. Finnish Prime Minister’s Office, 2022.
- [25] Ying Zhu and Katja Tähtinen. *Rakennusosien uudelleenkäytön edellytykset Suomessa*. Finnish Government’s analysis, assessment and research activities, 2022.